

Run 03 | NME

Global (including India)

IV only

Target Necessity-motivated entrepreneurship; binary 1 vs. 0
Sample 1,342,884 respondents **Coverage** 89 countries; 2015–2022
Positive cases 69,931 (5.21%) **Case time** 36m 15s
Train / val. / test 913,163 / 161,145 / 268,576 (68% / 12% / 20%, stratified)
Predictors (13): Self-efficacy, opportunity perception, fear of failure, knowing entrepreneurs, intention, media coverage, business exit, business angel, established business owner, good career choice, ease of starting a business, high status, social enterprise.
Fraser EFW predictors (7): Overall economic freedom, Economic freedom quartile, Size of government, Legal system and property rights, Sound money, Freedom to trade internationally, Regulation. Total independent variables: 20.
Controls (0): None. Country and respondent ID are excluded as predictors.
Protocol: Ten CPU models; model seed 20260925; fixed splits by sample/outcome; training-only preprocessing; F1 threshold selected on validation data. Test scores below are unweighted.

Result in brief

Highest observed ROC-AUC: Random Forest (0.9760). Highest F1: Random Forest (0.6312). The all-negative accuracy baseline is 94.79% (full sample).

Model performance on held-out test records

Model	ROC-AUC	PR-AUC	Acc.%	Prec.%	Recall%	F1	Time
Random Forest	0.9760	0.6800	95.88	59.17	67.62	0.6312	1m 47s
XGBoost	0.9759	0.6727	95.83	58.75	67.11	0.6265	18s
CatBoost	0.9750	0.6634	95.74	58.02	66.08	0.6179	1m 43s
H2O GBM	0.9735	0.6456	95.57	56.56	64.23	0.6015	58s
Decision Tree	0.9659	0.5787	94.40	47.31	66.41	0.5525	8s
Elastic Net	0.9640	0.4971	93.63	43.45	74.41	0.5487	10m 25s
Lasso	0.9640	0.4970	93.63	43.45	74.39	0.5486	9m 36s
Logistic Regression	0.9639	0.4958	93.18	41.88	79.84	0.5494	1m 25s
Ridge	0.9633	0.4854	93.14	41.70	79.53	0.5472	6m 55s
Naive Bayes	0.9006	0.2430	88.47	27.11	71.92	0.3938	2m 35s

Models ranked by ROC-AUC; small differences are descriptive, not significance tests. Time includes fit, prediction and output processing. Logistic regression uses a small stabilizing penalty; Lasso/Ridge/Elastic Net use 3-fold training CV.

Random-forest feature importance: top 8

Rank	Predictor	Raw importance	Share%
1	Entrepreneurial intention	18,156.07	36.19
2	Established business owner	6,861.48	13.68
3	Legal system and property rights	2,871.23	5.72
4	Size of government	2,663.37	5.31
5	Regulation	2,465.62	4.92
6	Freedom to trade internationally	2,446.14	4.88
7	Sound money	2,277.18	4.54
8	Overall economic freedom	2,175.13	4.34

Share is the percentage of total forest impurity importance across all 38 encoded predictors, not explained variance. Raw values are not comparable across samples. Missing indicators are derived preprocessing features.

Leading Fraser EFW predictors within the same forest

Rank	Fraser EFW predictor	Raw importance	Share%
1	Legal system and property rights	2,871.23	5.72
2	Size of government	2,663.37	5.31
3	Regulation	2,465.62	4.92

Fraser EFW subset ranked by raw importance; shares use total forest importance, not the Fraser EFW subtotal.

Source: Completed Objective 3 batch, 29 September 2026; 03_obj3_global_iv_NME. Scores and importance are taken directly from saved CSV outputs.